

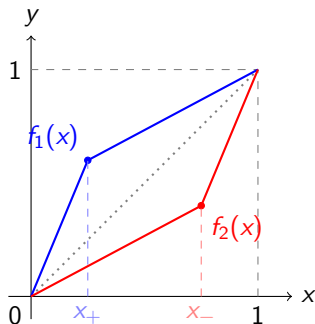
Unique ergodicity for random noninvertible maps on an interval

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Historical inspirations

L. Alsedà and M. Misiurewicz in *Random interval homeomorphisms*, *Publ. Mat.*, 58:15–36, 2014 studied the system of two transformations: f_1, f_2 selected randomly with probability $\mu(f_1) = \mu(f_2) = \frac{1}{2}$.



They looked for the existence of μ -invariant (also known as μ -stationary) ν on $(0, 1)$:

$$\nu(A) = \nu(f_1^{-1}A)\mu(f_1) + \nu(f_2^{-1}A)\mu(f_2) \quad \text{for every } A \in \mathcal{B}([0, 1]).$$

Introduction

Let μ be a Borel probability measure on the space $C([0, 1])$ of all continuous functions $g : [0, 1] \rightarrow [0, 1]$ equipped with the supremum norm $|\cdot|$. For brevity we will write gh for the composition $g \circ h$. By Γ we shall denote the topological support of μ .

Let $\mathbf{x}$ be a $[0, 1]$ -random variable, and let $\mathbf{g}$ be a Γ -random variable. Assume that $\mathbf{x}$ and $\mathbf{g}$ are independent, and that they have distributions ν and μ , respectively.

Then we easily see that the distribution of $\mathbf{g}(\mathbf{x})$ is equal to the convolution $\mu * \nu$ of the measures μ and ν defined by the formula:

$$\int_{[0,1]} f(x) d(\mu * \nu)(x) = \int_{\Gamma} \int_{[0,1]} f(g(x)) d\nu(x) d\mu(g)$$

for every bounded Borel measurable function $f : [0, 1] \rightarrow \mathbb{R}$. A Borel measure ν is said to be μ -invariant (or μ -stationary) if

$$\mu * \nu = \nu.$$

Introduction

A μ -invariant probability measure is said to be μ -ergodic if it cannot be written as a proper convex combination of two different μ -invariant probability measures.

The main purpose of this talk is to provide a condition on (Γ, μ) which implies unique ergodicity. An important component of this condition is the bounded variation of the random iteration, as follows:

(BV)

$$\mu^{\otimes \mathbb{N}}\left(\{(g_1, g_2, \dots) \in \Gamma^{\mathbb{N}} : \sup_{n \geq 1} \bigvee_0^1 g_1 g_2 \dots g_n < \infty\}\right) > 0,$$

where $\bigvee_0^1 g$ denotes the variation of the function g on $[0, 1]$ and $\mu^{\otimes \mathbb{N}}$ is the product measure on the product space $\Gamma^{\mathbb{N}}$.

Our main result is

Theorem 1

Let (Γ, μ) be a stochastic dynamical system on $C([0, 1])$ such that property (BV) is satisfied. Then the following statements hold:

- (i) If ν_1, ν_2 are μ -ergodic measures concentrated on $(0, 1)$ and such that

$$\{0, 1\} \cap \text{supp}(\nu_1) \cap \text{supp}(\nu_2) \neq \emptyset,$$

then $\nu_1 = \nu_2$.

- (ii) If (Γ, μ) has no atomic μ -invariant probability measures with finite support in the open interval $(0, 1)$, then any two distinct μ -ergodic measures ν_1 and ν_2 on $(0, 1)$ satisfy

$$\text{supp}(\nu_1) \cap \text{supp}(\nu_2) = \emptyset.$$

Recall that

$$\text{supp } \nu = \{x \in [0, 1] : \nu(B(x, r)) > 0 \text{ for all } r > 0\}.$$

Corollary 2

Let (Γ, μ) be a stochastic dynamical system on $C([0, 1])$ such that property (BV) is satisfied. Assume that

($\mathfrak{U}_1$) for every $x \in (0, 1)$ there exists $g \in \Gamma$ such that $g(x) < x$

or

($\mathfrak{U}_2$) for every $x \in (0, 1)$ there exists $g \in \Gamma$ such that $g(x) > x$.

Then, on $(0, 1)$, there is at most one μ -invariant probability measure. If additionally $\nu(\{0\}) = \nu(\{1\}) = 0$ for every μ -invariant probability measure, then (Γ, μ) is uniquely ergodic.

Corollary 3

Let (Γ, μ) be a stochastic dynamical system on $C([0, 1])$ such that property (BV) is satisfied. Assume additionally that both ($\mathfrak{U}_1$) and ($\mathfrak{U}_2$) hold. If $g((0, 1)) \subset (0, 1)$ for every $g \in \Gamma$ and there exist $g_0, g_1 \in \Gamma$ with $g_0(0) \in (0, 1)$ and $g_1(1) \in (0, 1)$, then (Γ, μ) is uniquely ergodic.

Examples

If all $g \in \Gamma$ are **monotonic**, then $\bigvee_0^1 g_1 \cdots g_n \leq 1$ for every $g_1, \dots, g_n \in \Gamma$ and condition (BV) holds. If we additionally assume that both $(\mathfrak{U}_1)$ and $(\mathfrak{U}_2)$ are satisfied, then, according to Corollary 2, (Γ, μ) has at most one μ -invariant measure, which is concentrated on $(0, 1)$. Moreover, if there exist $g_0, g_1 \in \Gamma$ with $g_0(0) \in (0, 1)$ and $g_1(1) \in (0, 1)$, then (Γ, μ) is uniquely ergodic, by Corollary 3.

In [BOS] we introduced the so-called **μ -injective maps**. Recall that the stochastic dynamical system (Γ, μ) on $C([0, 1])$ is said to be *μ -injective* if for every $x \in [0, 1]$,

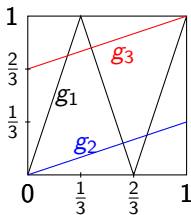
$$\int_{\Gamma} \#g^{-1}(x) \mu(dg) \leq 1.$$

We proved that every μ -injective system (Γ, μ) on $C([0, 1])$ satisfies condition (BV). If, additionally, the hypotheses of Corollary 2 or 3 hold, then the system has at most one μ -invariant probability measure on $(0, 1)$ or is uniquely ergodic.

Example 4

Let us consider the following three continuous functions.

$$g_1(x) := \begin{cases} 3x & \text{if } x \in [0, \frac{1}{3}], \\ -3x + 2 & \text{if } x \in (\frac{1}{3}, \frac{2}{3}], \\ 3x - 2 & \text{if } x \in (\frac{2}{3}, 1], \end{cases} \quad g_2(x) := \frac{x}{3} \quad \text{and} \quad g_3(x) := \frac{x}{3} + \frac{2}{3}.$$



Let us take $\Gamma = \{g_1, g_2, g_3\}$ and $\mu(\{g_2\}) = \mu(\{g_3\}) =: p$. Since $\#g_1^{-1}(\{x\}) = 3$ for $0 < x < 1$, we see that the system is μ -injective for the case that $p \geq \frac{2}{5}$ and not μ -injective otherwise.

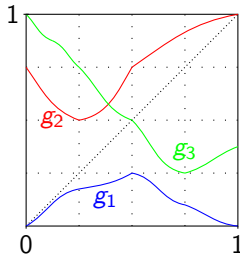
Moreover, this system has additional features, namely it satisfies conditions $(\mathfrak{U}_1)$ and $(\mathfrak{U}_2)$.

Systems constructed by tableaux. Let $\mathcal{I} = \{I_1, \dots, I_m\}$ be a finite covering of $[0, 1]$ by closed intervals such that for every $g \in \Gamma$ and for every $I \in \mathcal{I}$:

- (1) g is monotone on I ,
- (2) there exists $J \in \mathcal{I}$ such that $g(I) \subset J$.

Then for any probability measure μ on $C([0, 1])$ with $\text{supp}(\mu) = \Gamma$, condition (BV) is satisfied.

Let us consider the discrete system $\Gamma = \{g_1, g_2, g_3\}$ defined by the drawing below.



Systems constructed by tableaux — uniquely ergodic by Corollary 3

μ -injective systems — stability

Let P be a Markov operator acting on Borel measures on $[0, 1]$ corresponding to the system (Γ, μ) , i.e.

$$P\nu = \mu * \nu.$$

The system is said to be **asymptotically stable** if it is uniquely ergodic and $(P^n \nu)_{n \geq 1}$, for any probability measure ν , converges weakly to the μ -invariant measure ν_* .

Define

$(\mathfrak{U}_1^+)$ there exists $g \in \Gamma$ such that $g(x) < x$ for every $x \in (0, 1]$
and

$(\mathfrak{U}_2^+)$ there exists $g \in \Gamma$ such that $g(x) > x$ for every $x \in [0, 1)$.

Theorem 5

Let (Γ, μ) be μ -injective, and let conditions $(\mathfrak{U}_1^+)$ and $(\mathfrak{U}_2^+)$ hold. Then its corresponding Markov operator is asymptotically stable.

μ -injective systems — proximality

It is well known that the sequence of measures $(\nu(g_n^{-1} \circ \dots \circ g_1^{-1}(\cdot)))_{n \geq 1}$ is weakly convergent for $\mu^{\otimes \mathbb{N}}$ -a.e. $\omega = (g_1, g_2, \dots) \in \Gamma^{\mathbb{N}}$ (see [F], Corollary p. 20). Let us denote its limit by ν_ω . The measures ν will satisfy the barycenter equation

$$\nu(\cdot) = \int_{\Gamma^{\mathbb{N}}} \nu_\omega(\cdot) \mu^{\otimes \mathbb{N}}(d\omega).$$

Definition 1 (H. Furstenberg)

Following Furstenberg we say that a μ -invariant measure ν for a given stochastic dynamical system (Γ, μ) is μ -proximal (or just *proximal*) if there exists a measurable function $\Theta : \Gamma^{\mathbb{N}} \rightarrow [0, 1]$ such that

$$\nu_\omega = \delta_{\Theta(\omega)}$$

for $\mu^{\otimes \mathbb{N}}$ -a.e. $\omega = (g_1, g_2, \dots) \in \Gamma^{\mathbb{N}}$.

μ -injective systems — proximality

Theorem 6

Let (Γ, μ) be μ -injective, and let conditions $(\mathfrak{U}_1^+)$ and $(\mathfrak{U}_2^+)$ hold. Then the unique μ -invariant measure ν_* is proximal.

Remark 7

Note that if we know that any μ -invariant measure is proximal, then it must be unique. Indeed, if we have two $\nu^1, \nu^2 \in \mathcal{P}$, then we have $\nu_\omega^i = \delta_{\Theta_i(\omega)}$, $i = 1, 2$, for $\mu^{\otimes \mathbb{N}}$ -a.e. ω and some measurable functions $\Theta_1, \Theta_2 : \Gamma^{\mathbb{N}} \rightarrow [0, 1]$. Since $\nu^i = \int_{\Gamma^{\mathbb{N}}} \nu_\omega^i d\mu^{\otimes \mathbb{N}}$ for $i = 1, 2$ and $\nu^1 \neq \nu^2$, we obtain

$$\mu^{\otimes \mathbb{N}}(\{\omega \in \Gamma^{\mathbb{N}} : \Theta_1(\omega) \neq \Theta_2(\omega)\}) > 0. \quad (1)$$

On the other hand, $\nu^3 = (\nu^1 + \nu^2)/2$ is μ -invariant and μ -proximal. Then $\nu_\omega^3 = \delta_{\Theta_3(\omega)}$ for $\mu^{\otimes \mathbb{N}}$ -a.e. ω and some measurable functions $\Theta_3 : \Gamma^{\mathbb{N}} \rightarrow [0, 1]$. The easy observation that

$$\nu_\omega^3 = \frac{\nu_\omega^1 + \nu_\omega^2}{2} = \frac{\delta_{\Theta_1(\omega)} + \delta_{\Theta_2(\omega)}}{2} \neq \delta_{\Theta_3(\omega)}$$

on some $\mu^{\otimes \mathbb{N}}$ -positive subset of $\omega \in \Gamma^{\otimes \mathbb{N}}$, by (1), leads to a contradiction

μ -injective systems — central limit theorem

We have

if $\xi_0 = x$ then $\xi_n(\omega) = g_n \circ \cdots \circ g_1(x)$ for $\omega = (g_1, g_2, \dots) \in \Gamma^{\mathbb{N}}$

Finally, we prove that if (Γ, μ) is μ -injective, then the stochastic process $(\varphi(\xi_n))_{n \geq 0}$, where $(\xi_n)_{n \geq 0}$ denotes the corresponding stationary Markov chain with initial distribution ν (ν is a μ -invariant measure) and φ is a *centered Lipschitz function* on $[0, 1]$, i.e., $\int_{[0,1]} \varphi(x) \nu(dx) = 0$, satisfies the *Annealed Central Limit Theorem*, i.e.,

$$\sigma^2 := \lim_{n \rightarrow \infty} \mathbb{E}_{\nu} \left(\frac{\varphi(\xi_1) + \cdots + \varphi(\xi_n)}{\sqrt{n}} \right)^2$$

exists and

$$\frac{\varphi(\xi_1) + \cdots + \varphi(\xi_n)}{\sqrt{n}} \Rightarrow \mathcal{N}(0, \sigma^2) \quad \text{as } n \rightarrow \infty,$$

where $\Rightarrow$ denotes convergence in distribution.

μ -injective systems — central limit theorem

Theorem 8

Let (Γ, μ) be μ -injective, and let conditions $(\mathfrak{U}_1^+)$ and $(\mathfrak{U}_2^+)$ hold. Let $(\xi_n)_{n \geq 0}$ be the corresponding stationary Markov chain with initial distribution ν . Then for any centered Lipschitz function $\varphi : [0, 1] \rightarrow \mathbb{R}$ the random process $(\varphi(\xi_n))_{n \geq 1}$ satisfies the central limit theorem. Moreover, the same is true for the process $(\varphi(\xi_n^x))_{n \geq 1}$, where $(\xi_n^x)_{n \geq 1}$ is the corresponding Markov chain starting from an arbitrary point $x \in [0, 1]$.



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